How To Run

cd /where-you-downloaded-the-thing-to/A3

python3 -m pip install -r requirements.txt

cd /where-you-downloaded-the-thing-to/A3/messaging_service

python3 app.py

Example of running:

```
moonbox@bluemoonshomepc:~/school/Notes$ cd A3/
moonbox@bluemoonshomepc:~/school/Notes/A3$ python3 -m pip install -r requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: flask in /home/moonbox/.local/lib/python3.10/site-packages
 (from -r requirements.txt (line 1)) (3.1.3)
Requirement already satisfied: flask-cors in /home/moonbox/.local/lib/python3.10/site-pac
kages (from -r requirements.txt (line 2)) (6.0.2)
Requirement already satisfied: blinker>=1.9.0 in /home/moonbox/.local/lib/python3.10/site
-packages (from flask->-r requirements.txt (line 1)) (1.9.0)
Requirement already satisfied: click>=8.1.3 in /home/moonbox/.local/lib/python3.10/site-p
ackages (from flask->-r requirements.txt (line 1)) (8.3.1)
Requirement already satisfied: itsdangerous>=2.2.0 in /home/moonbox/.local/lib/python3.10
/site-packages (from flask->-r requirements.txt (line 1)) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in /usr/local/lib/python3.10/dist-packages (
from flask->-r requirements.txt (line 1)) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.10/dist-packag
es (from flask->-r requirements.txt (line 1)) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in /home/moonbox/.local/lib/python3.10/sit
e-packages (from flask->-r requirements.txt (line 1)) (3.1.6)
moonbox@bluemoonshomepc:~/school/Notes/A3$ cd messaging_service/
moonbox@bluemoonshomepc:~/school/Notes/A3/messaging_service$ python3 app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a pr
oduction WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

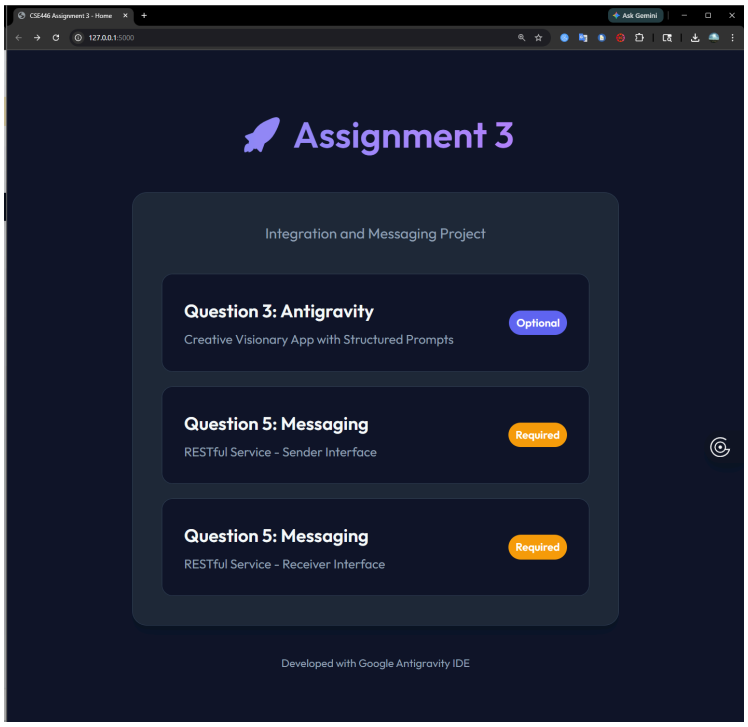
node No...
node A3
python3...

Your application running on port 5000 is available. See all forwarded ports

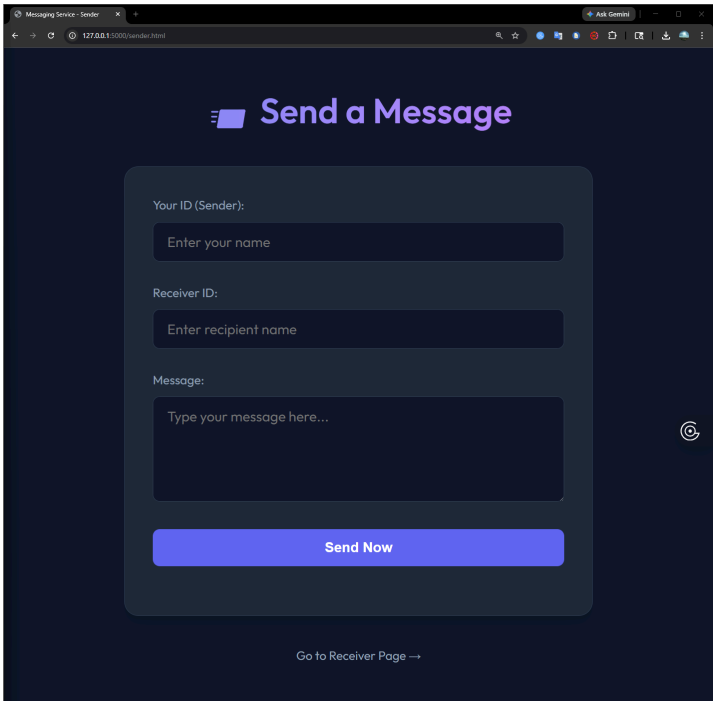
Open in Browser | Preview in Editor

Screenshot proofs:

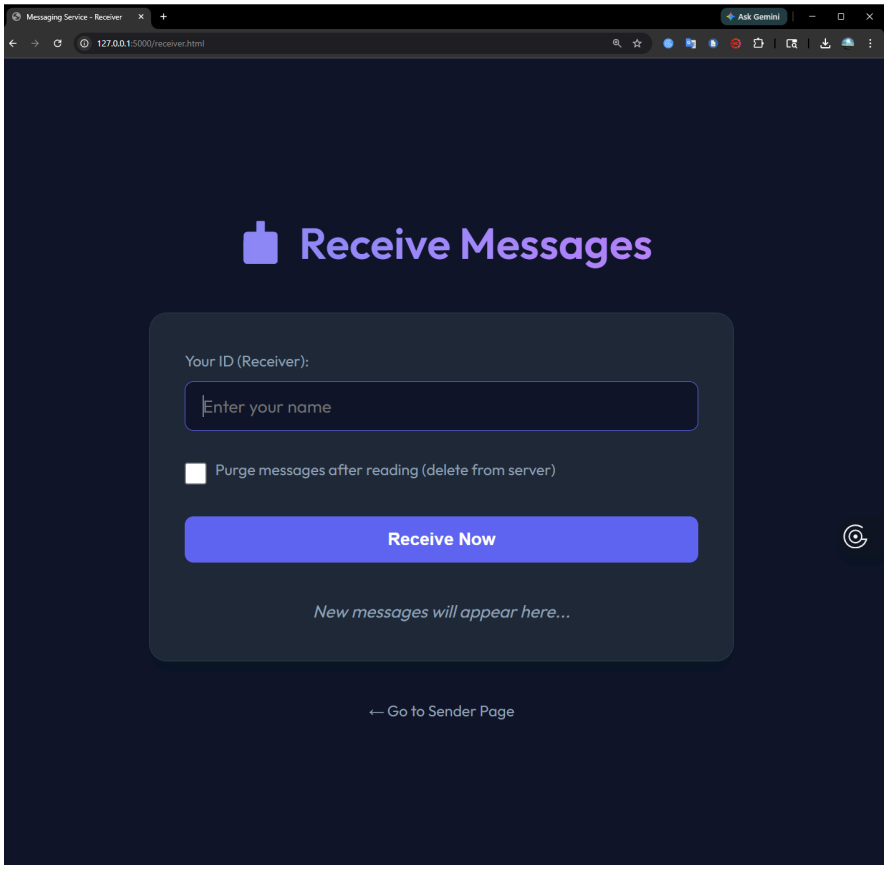
1. Service running in browser at http://localhost:5000/ (home page visible).



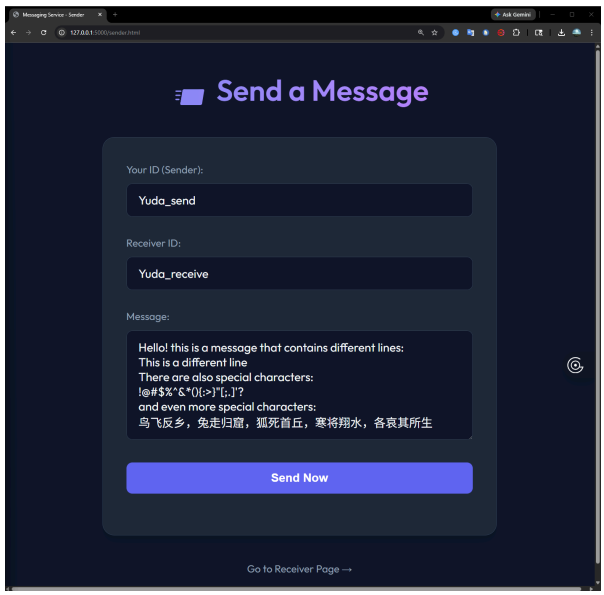
2. Sender page open at http://localhost:5000/sender.html with filled fields before send.

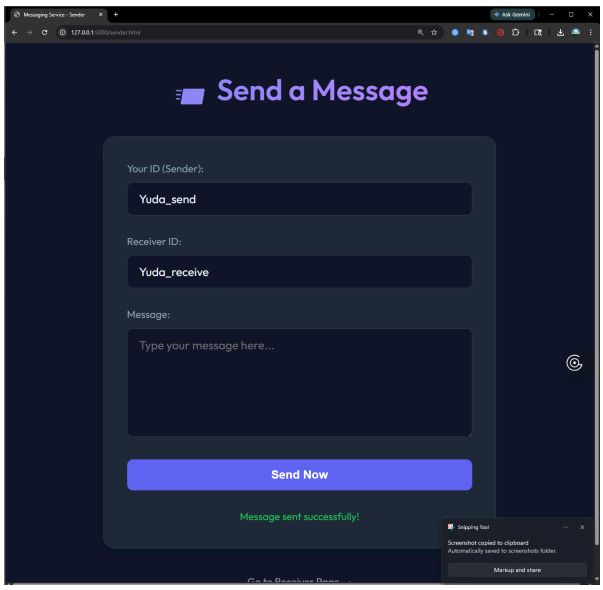


3. Receiver page open at http://localhost:5000/receiver.html with receiver ID before receive.

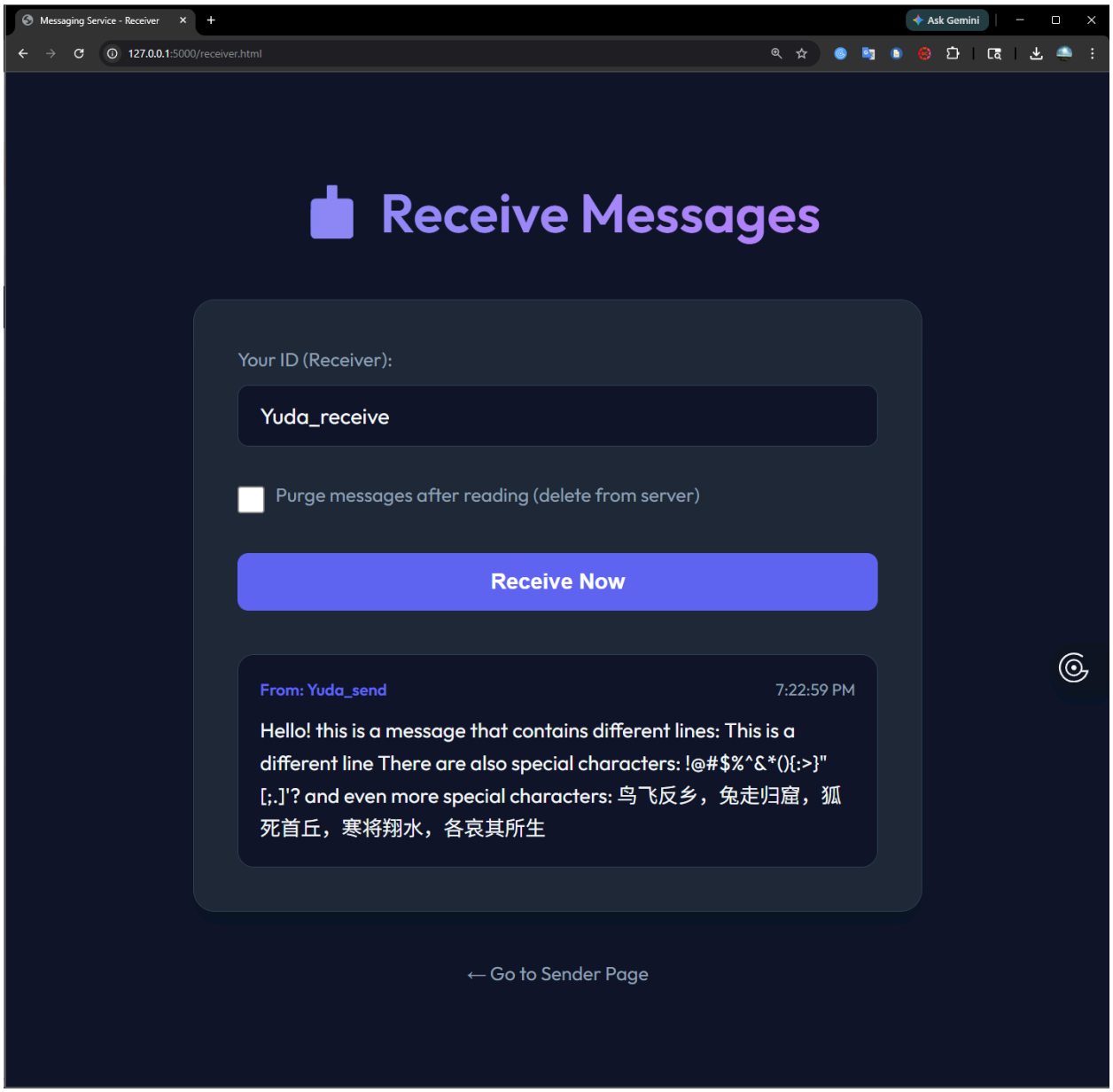


4. After clicking send: sender page showing success message.

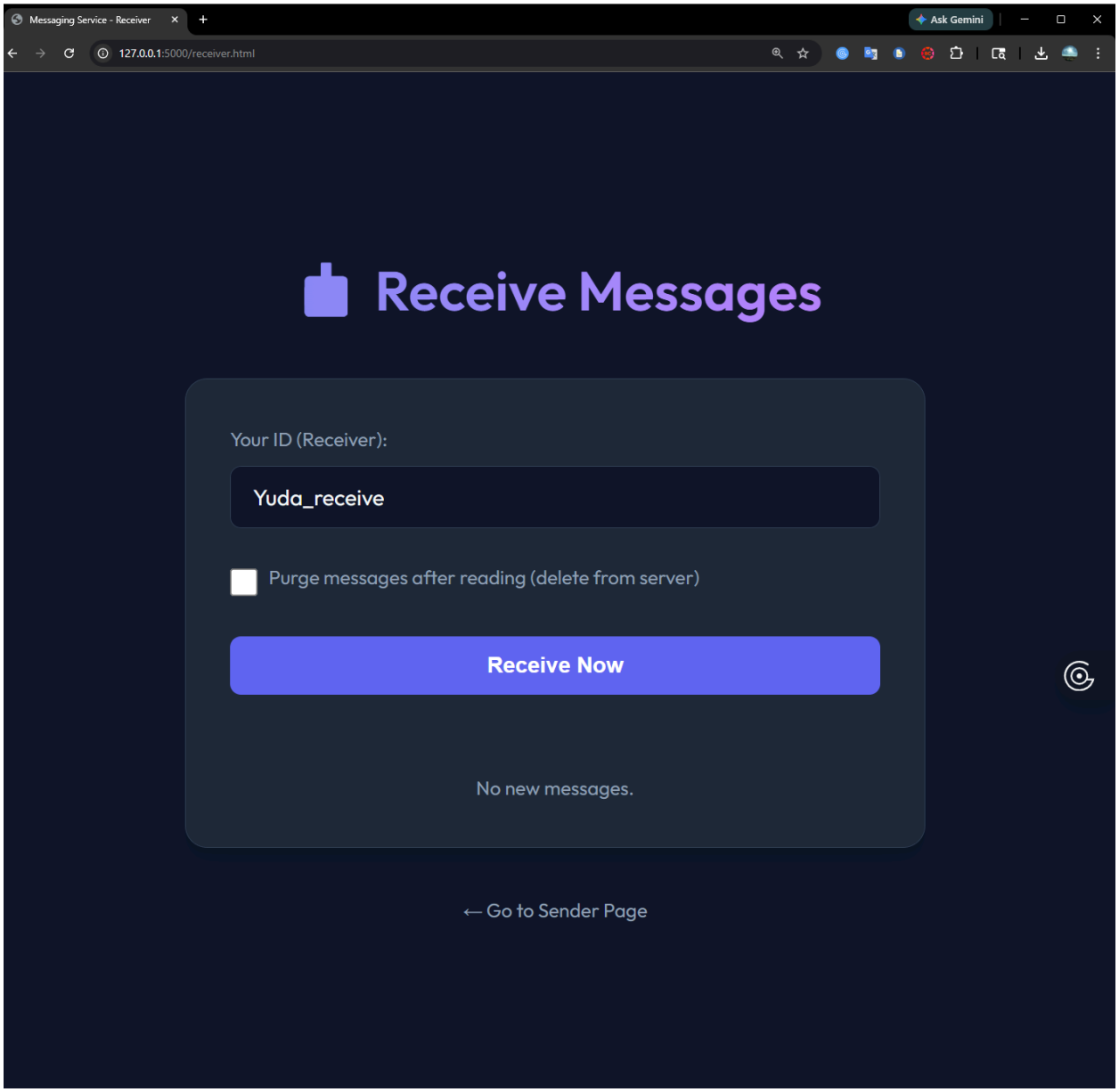




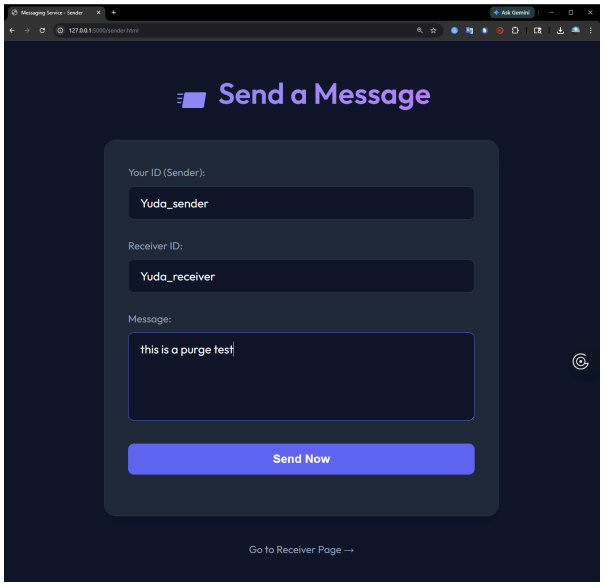
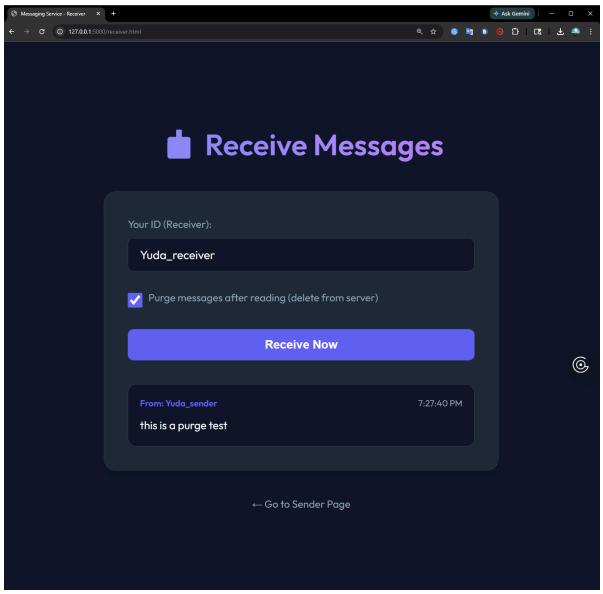
5. Receiver after first receive: message card visible with sender/time/message.

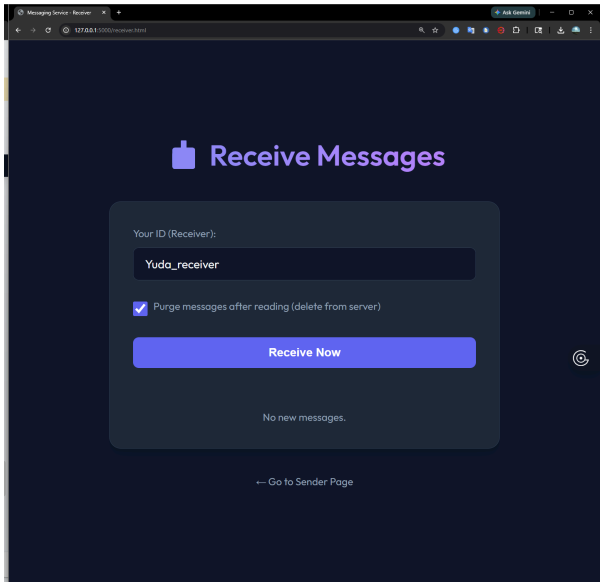


6. Receiver second receive (without new send): showing “No new messages.”


Messaging Service - Receiver
Ask Gemini
127.0.0.1:5000/receiver.html
Receive Messages
Your ID (Receiver):
Yuda_receive
Purge messages after reading (delete from server)
Receive Now
No new messages.
← Go to Sender Page

7. Purge test:

Send another message.	
Receiver with purge checked gets it once.	

Next receiver shows “No new messages.”	

8. Terminal screenshot showing Flask app running

```
moonbox@bluemoonshomepc:~/school/Notes/A3/messaging_service$ python3 app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a pr
oduction WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [01/Mar/2026 19:17:54] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:17:54] "GET /style.css HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:17:55] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [01/Mar/2026 19:18:19] "GET /sender.html HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:18:19] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:19:57] "GET /receiver.html HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:19:57] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:20:46] "GET /sender.html HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:20:46] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:22:59] "POST /sendMsg HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:24:29] "GET /receiver.html HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:24:29] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:24:59] "GET /receiveMsg?receiverID=Yuda_receive&purge=false
 HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:25:19] "GET /receiveMsg?receiverID=Yuda_receive&purge=false
 HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:26:30] "GET /sender.html HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:26:30] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:27:00] "POST /sendMsg HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:27:01] "GET /receiver.html HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:27:01] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:27:13] "GET /sender.html HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:27:13] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:27:40] "POST /sendMsg HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:27:44] "GET /receiver.html HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:27:44] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:27:55] "GET /receiveMsg?receiverID=Yuda_receiver&purge=true
 HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 19:28:07] "GET /sender.html HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:28:07] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:28:09] "GET /receiver.html HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:28:09] "GET /style.css HTTP/1.1" 304 -
127.0.0.1 - - [01/Mar/2026 19:28:23] "GET /receiveMsg?receiverID=Yuda_receiver&purge=true
 HTTP/1.1" 200 -
```

Implemented pages:

The 3 browser windows:

service/home:

- http://localhost:5000/

Sender:

- http://localhost:5000/sender.html

Receiver:

- http://localhost:5000/receiver.html